# VEDULA ABHISHEK

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## SUMMARY

I am a postgraduate student currently pursuing M. Tech in Computer Science (AI) with strong interest in Data Science and Analytics. I have worked on projects involving data cleaning, analysis, visualization, and machine learning models using Python, SQL, and Excel. I enjoy finding patterns in data and creating insights that support decision making. As a fresher, I am eager to learn from industry projects, grow my skills, and start my career in Data Science.

## EDUCATION

**Centre for Development of Advanced Computing (C-DAC, Mohali)**

**Master of Technology (M. Tech):** Computer Science and Engineering (AI)
CGPA: 8.72, 2024-2026 (Expected)

**Bhilai Institute of Technology (BIT Raipur)**

**Bachelor of Technology (B. Tech):** Computer Science and Engineering
CGPA: 8.8, 2019-2023

## WORK EXPERIENCE

**Data science intern at AI Variant** December 2023 – June

- Streamlined data preprocessing and analytics pipelines, reducing model deployment time and improving operational efficiency.
- Designed and deployed production-grade ML models using Python, Pandas, and Streamlit for real-time analysis and visualization
- Collaborated with team members to fine-tune algorithms, resulting in higher accuracy and better generalization.
- Performed comprehensive EDA on large datasets, identifying patterns and insights to guide strategic decisions.
- Prepared stakeholder-ready reports and presentations, ensuring complex technical findings were communicated clearly and effectively.

## PROJECT

**Brain Tumor Prediction (Deep Learning Model):**

- Objective: To build a highly accurate machine learning model for the detection of brain tumors from medical image data.
- Methodology: Collected and processed a diverse dataset of images. Conducted extensive data analysis to identify key features and trends. Tested and optimized various classification algorithms to maximize model performance
- Result: Developed a model that achieved high accuracy in predicting brain tumors. The project was showcased via a live web application built with Streamlit.
- Tech: Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, CNN, Deep Learning
- Link: GitHub | Streamlit

**AI Medical Q&A Chatbot (RAG-based Information Retrieval System):**

- Objective: To provide accurate, multilingual answers to disease-related queries by retrieving structured medical information from Wikipedia
- Methodology: Extracted key medical sections (Symptoms, Causes, Treatment, etc.) from Wikipedia, generated Q&A pairs, and implemented semantic similarity search using SentenceTransformers for best-match retrieval. Integrated multilingual support via GoogleTranslator and built an interactive Streamlit UI for real-time query handling.
- Result: Delivered precise, context-aware medical answers with matched question reference and confidence scoring, enhancing user trust and accessibility.
- Tech: Python, Streamlit, SentenceTransformers, Wikipedia API, GoogleTranslator, PyTorch
- Link: GitHub

**Book Recommendation System:**

- Objective: To develop a personalized book recommendation system that suggests relevant titles to users.
- Methodology: Employed collaborative filtering techniques to analyze user ratings and identify users with similar tastes. The system was built using Python, with Pandas for data manipulation and scikit-learn for the recommendation engine.
- Result: Deployed the model as an interactive web application using Streamlit, providing users with a dynamic platform to receive personalized and accurate book suggestions.
- Tech: Python, Pandas, Scikit-learn, Streamlit, Numpy, clustering
- Link: GitHub

## TECHNICAL SKILLS

- **Programming:** Python, SQL, API Integration
- **AI/ML Expertise:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Optimization, Time Series Forecasting
- **NLP & Generative AI:** Named Entity Recognition, Embeddings, Summarization, RAG Pipelines, Prompt Engineering (Hugging Face, Lang Chain)
- **Computer Vision:** OpenCV, CNNs, YOLO, RetinaNet, Segmentation, OCR
- **Frameworks & Libraries:** Scikit-learn, TensorFlow/Keras, Transformers, Pandas, NumPy, Matplotlib, Seaborn
- **Cloud & Deployment:** Azure AI Studio (Basics), Streamlit Cloud, Google Colab, Docker
- **Tools & Platforms:** Jupyter Notebook, Power BI, Excel, GitHub, Google Workspace (Docs, Sheets, Gmail)

## CERTIFICATES

- **Data Science Certification** - Excelr
- **SQL Intermediate** - HackerRank
- **Master's Program in Data Science** - NASSCOM
- **Introduction to Python** - Coding Ninjas